NON-WSQ COURSE SLIDES · DESIGN OF EXPERIMENTS

Design of Experiment (DOE) Masterclass

Course Code: C950
2 days · 15 instructional hours
Trainer: Dr Alfred Ang



Tertiary Infotech Academy Pte Ltd · UEN 201200696W
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COURSE ADMINISTRATION

Welcome & Course Orientation

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About the Trainer



Your Trainer

Trainer must complete this profile before delivery

NAME _____

TITLE / DESIGNATION _____

QUALIFICATIONS _____

EXPERTISE _____

EXPERIENCE _____

CONTACT _____

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NAMED TRAINER

About the Trainer



Dr Alfred Ang

Principal Trainer · Tertiary Infotech Academy

ROLE	Principal Trainer and Chief Instructional Designer
EXPERTISE	Applied research, engineering analytics and experimental design
CERTIFICATION	ACLP-certified adult educator
DELIVERS	Engineering, data and continuous-improvement programmes
EXPERIENCE	Cross-industry experimentation and workforce training
CONTACT	enquiry@tertiaryinfotech.com

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LEARNER INTRODUCTION

Let's Know Each Other

1

Your work

Role, organisation and process or research system.

2

Your DoE baseline

ANOVA, factorials, Minitab, Excel or no prior experience.

3

Your experiment

One response you want to improve or stabilise.

4

Your outcome

A decision you want stronger evidence to support.

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HOUSEKEEPING

Ground Rules

1. **Be present**
   Return from breaks on time — attend both days to get the most from the course.

2. **Participate**
   Ask, test and explain your reasoning.

3. **Respect**
   One conversation; critique ideas, not people.

4. **Work safely**
   Use only feasible factor ranges and supplied mock data.

5. **Preserve evidence**
   Keep standard order, run order and source responses.

6. **Learn by doing**
   Use the Learner Guide for detailed procedures.

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SCHEDULE

Lesson Plan — 2 Days

DAY 1

1. 9:30–10:55 · DoE foundations and project framing
2. 10:55–12:45 · Hypothesis testing and one-way ANOVA
3. 1:15–3:05 · 2^2 and 2^3 factorial designs
4. 3:05–5:30 · Effects, interactions, regression and blocks

DAY 2

1. 9:30–11:25 · Fractional factorial and aliases
2. 11:25–2:25 · Screening and sequential design
3. 2:25–4:15 · RSM, CCD, mixtures and Taguchi
4. 4:15–5:30 · Evaluation, recap and course feedback

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COURSE REPOSITORY

Access the NovaDOE Hands-On Labs

Option A · Clone

- Use Git to clone the repository
- Open the numbered DOE lab folder
- Start with README and workbook

Option B · Download ZIP

- GitHub > Code > Download ZIP
- Extract the full activities tree
- Keep each lab self-contained

NovaDOE workflow

- Generate the design in the browser
- Export CSV into the lab folder
- Use the LG for detailed procedures

https://github.com/tertiarycourses/C950-Design-of-Experiment-DOE-Masterclass

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Download Course Material

1 · Sign in

- Open the LMS/TMS portal
- Use your learner account
- Select this course

2 · Download

- Open Courseware
- Download learner slides and guide
- Save beside your lab folders

3 · Verify

- **Check the course code**
- Use the current version
- Ask the trainer if a file differs

https://lms-tms.tertiaryinfotech.com/

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01

TOPIC 01

Fundamentals of Design of Experiments

Questions, variables, variation, hypothesis tests and one-way ANOVA

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ALIGNMENT · LO1

Key Concepts — Fundamentals of Design of Experiments

1. **Engineering experiments**
   Learn efficiently from controlled variation

2. **DoE lifecycle**
   Plan before collecting data

3. **Factor, level, treatment, response**
   Name the experimental structure

4. **Independent and dependent variables**
   Separate inputs from outcomes

5. **OFAT versus DoE**
   Interactions require coordinated changes

6. **Randomisation, replication and blocking**
   Protect inference from bias and noise

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Engineering experiments

1. **Reduce** development time
2. **improve** performance
3. **quantify** uncertainty
4. **build** robust products

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Engineering experiments — Decision Lens

INTERPRET

1. Learn efficiently from controlled variation
2. Evidence question: What observation would change the current decision?
3. Model risk: Check assumptions before interpreting the effect.

APPLY

1. A useful experiment changes a decision, not merely a chart
2. Engineering use: Translate coded results back to operating units.
3. Next action: Confirm the recommendation in an independent run.

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DoE lifecycle

State

- problem

choose

- factors and levels

select

- response

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DECISION LENS · LO1

DoE lifecycle — Decision Lens

1

Meaning

Plan before collecting data

2

Evidence

Identify the observation that would change the decision.

3

Guardrail

Pre-register the response and decision rule

4

Follow-up

Confirm the recommendation independently.

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Factor, level, treatment, response

1

Factor — is controlled input

2

level — is setting

3

treatment — is combination

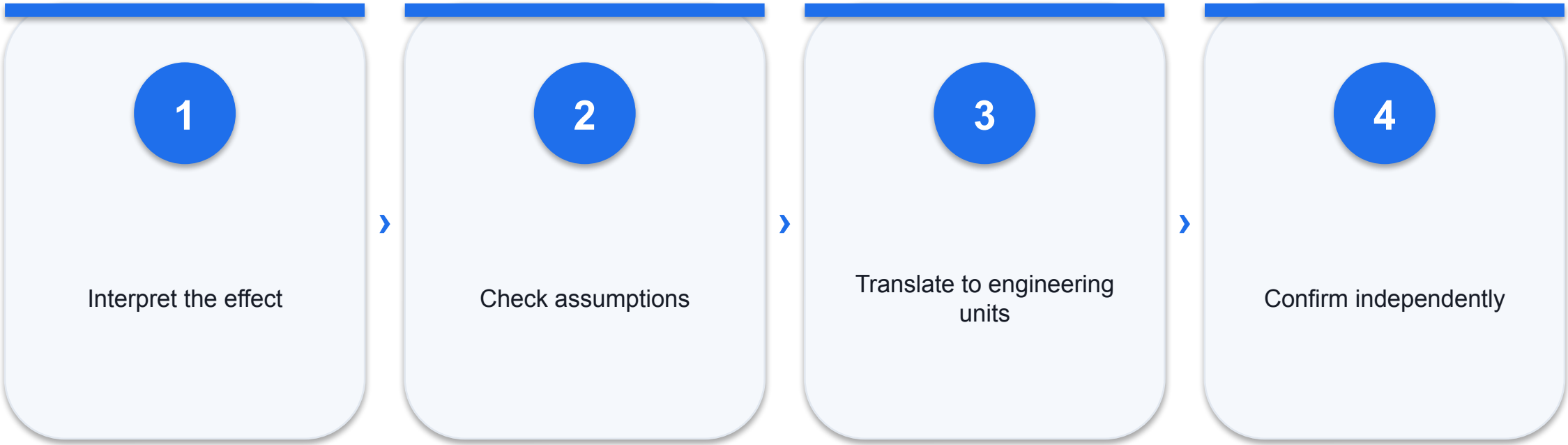
4

response — is measured output

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DECISION LENS · LO1

Factor, level, treatment, response — Decision Lens



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Independent and dependent variables

STRUCTURE

1. Independent variables are manipulated
2. dependent variables respond

EVIDENCE

1. nuisance variables are controlled or blocked

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Independent and dependent variables — Decision Lens

Interpret

- Separate inputs from outcomes
- State the evidence needed.

Control

- Do not call an observed association an intervention
- Check model assumptions.

Act

- Translate to operating units.
- Confirm in an independent run.

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OFAT versus DoE

1. **OFAT**
   is intuitive but inefficient

2. **factorial**
   DoE estimates several effects

3. **interactions**
   are visible

4. **prediction**
   covers the region

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OFAT versus DoE — Decision Lens

INTERPRET

1. Interactions require coordinated changes
2. Evidence question: What observation would change the current decision?
3. Model risk: Check assumptions before interpreting the effect.

APPLY

1. Use OFAT only for safe preliminary range-finding
2. Engineering use: Translate coded results back to operating units.
3. Next action: Confirm the recommendation in an independent run.

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Randomisation, replication and blocking

Randomise

- run order

replicate

- treatments

block

- known nuisance sources

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DECISION LENS · LO1

Randomisation, replication and blocking — Decision Lens

1

Meaning

Protect inference from bias and noise

2

Evidence

Identify the observation that would change the decision.

3

Guardrail

Repeats measure equipment precision; replicates estimate process variation

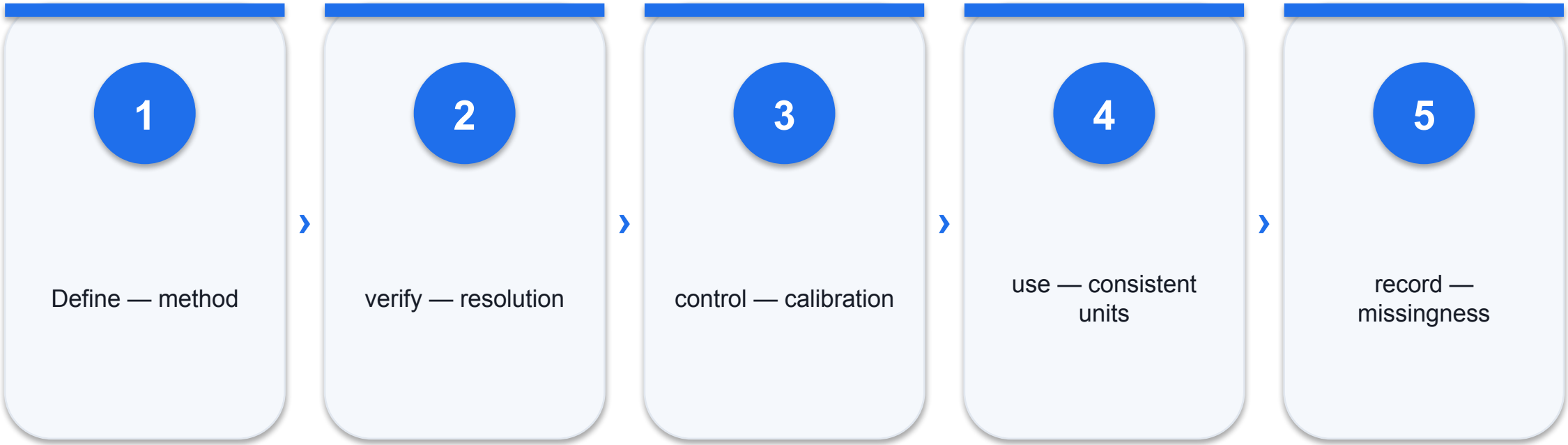
4

Follow-up

Confirm the recommendation independently.

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Measurement systems



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Measurement systems — Decision Lens

1. Interpret the effect
2. Check assumptions
3. Translate to engineering units
4. Confirm independently

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Hypothesis testing

STRUCTURE

1. Write null and alternative
2. choose statistic

EVIDENCE

1. calculate p-value
2. compare with alpha

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Hypothesis testing — Decision Lens

Interpret

- Translate evidence into a decision rule
- State the evidence needed.

Control

- A p-value is conditional on the null and assumptions
- Check model assumptions.

Act

- Translate to operating units.
- Confirm in an independent run.

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Type I and Type II errors

Type

- I rejects a true null

Type

- II misses a real effect

power

- depends on effect, noise and sample size

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DECISION LENS · LO1

Type I and Type II errors — Decision Lens

INTERPRET

1. Balance false alarms and missed effects
2. Evidence question: What observation would change the current decision?
3. Model risk: Check assumptions before interpreting the effect.

APPLY

1. Choose risk thresholds from consequence, not habit
2. Engineering use: Translate coded results back to operating units.
3. Next action: Confirm the recommendation in an independent run.

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Two-sample comparisons

Independent

- groups differ from paired measurements

inspect

- variance

report

- effect and uncertainty

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DECISION LENS · LO1

Two-sample comparisons — Decision Lens

1

Meaning

Match the test to the sampling design

2

Evidence

Identify the observation that would change the decision.

3

Guardrail

Statistical significance does not guarantee practical value

4

Follow-up

Confirm the recommendation independently.

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One-way ANOVA

1. Partition — total variation
2. estimate — between and within variation
3. compute — F ratio
4. test — equal means

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One-way ANOVA — Decision Lens

1. Interpret the effect
2. Check assumptions
3. Translate to engineering units
4. Confirm independently

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Sum-of-squares decomposition

STRUCTURE

1. SST equals model plus error
2. degrees of freedom scale mean squares

EVIDENCE

1. F compares signal to noise

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Sum-of-squares decomposition — Decision Lens

Interpret

- Account for where variation comes from
- State the evidence needed.

Control

- Balanced designs make decomposition easier to audit
- Check model assumptions.

Act

- Translate to operating units.
- Confirm in an independent run.

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TOPIC 01 · HANDS-ON

Characterise a Composite Cure Process

LAB 1

A materials team must reduce cure-cycle variation before testing a new composite panel.

EXPECTED OUTPUT

A signed experiment charter, factor table, 2^2 design CSV and run-order evidence.

TOOLS NovaDOE · NovaDOE full factorial planning workspace · Excel/CSV

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Characterise a Composite Cure Process

ACCEPTANCE CHECK

The factor table, four treatment combinations, response definition, random run order and go/no-go rule are all present.

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TOPIC 01 · HANDS-ON

Compare Coating Cure Temperatures with One-Way ANOVA

LAB 2

A coating line compares four cure temperatures using adhesion strength from replicated coupons.

EXPECTED OUTPUT

A randomised 4-level design, replicated response dataset, ANOVA table and engineering interpretation.

TOOLS NovaDOE · NovaDOE full factorial (one 4-level categorical factor) + Excel ANOVA · Excel/CSV

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Compare Coating Cure Temperatures with One-Way ANOVA

ACCEPTANCE CHECK

The ANOVA uses 20 observations, reports the correct factor levels and gives a conclusion that separates statistical from practical significance.

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TOPIC RECAP

Recap — Fundamentals of Design of Experiments

1

Knowledge

Engineering experiments and related principles.

2

Ability

Frame a measurable DoE project and distinguish factors, levels, responses and nuisance variables.

3

Evidence

A NovaDOE export, validated workbook and engineering interpretation.

4

Next

Use the evidence to narrow, augment or confirm the design.

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02

TOPIC 02

Factorial Design of Experiments

Main effects, interactions, regression, ANOVA, blocks and confounding

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ALIGNMENT · LO2

Key Concepts — Factorial Design of Experiments

1. **Factorial design**
   Test combinations rather than isolated changes

2. **The 2^2 design**
   Two factors at low and high levels

3. **Main effects**
   Average response change across another factor

4. **Interactions**
   The effect of one factor depends on another

5. **Coded regression model**
   Connect design geometry to prediction

6. **ANOVA for factorial designs**
   Test model terms against residual noise

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Factorial design

All

- combinations estimate main effects

interaction

- estimates become possible

one

- dataset supports prediction

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Factorial design — Decision Lens

INTERPRET

1. Test combinations rather than isolated changes
2. Evidence question: What observation would change the current decision?
3. Model risk: Check assumptions before interpreting the effect.

APPLY

1. Keep the experimental region safe and relevant
2. Engineering use: Translate coded results back to operating units.
3. Next action: Confirm the recommendation in an independent run.

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The 2^2 design

Four

- treatment combinations

coded

- levels -1 and +1

geometric

- square

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The 2^2 design — Decision Lens

1. **Meaning**
   Two factors at low and high levels

2. **Evidence**
   Identify the observation that would change the decision.

3. **Guardrail**
   Add replicates or centre points when pure error or curvature matters

4. **Follow-up**
   Confirm the recommendation independently.

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Main effects

1. Compare — high versus low means
2. sign — gives direction
3. magnitude — gives practical influence
4. uncertainty — qualifies the estimate

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Main effects — Decision Lens

1. Interpret the effect
2. Check assumptions
3. Translate to engineering units
4. Confirm independently

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Interactions

STRUCTURE

1. Non-parallel interaction lines
2. product term in regression

EVIDENCE

1. hierarchy principle
2. joint settings drive the decision

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Interactions — Decision Lens

Interpret

- The effect of one factor depends on another
- State the evidence needed.

Control

- Never interpret main effects alone when interaction is material
- Check model assumptions.

Act

- Translate to operating units.
- Confirm in an independent run.

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Coded regression model

1. **Intercept**
   is grand mean

2. **coefficients**
   represent half-effects

3. **product**
   term models interaction

4. **residuals**
   test adequacy

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DECISION LENS · LO2

Coded regression model — Decision Lens

INTERPRET

1. Connect design geometry to prediction
2. Evidence question: What observation would change the current decision?
3. Model risk: Check assumptions before interpreting the effect.

APPLY

1. Decode recommendations back to engineering units
2. Engineering use: Translate coded results back to operating units.
3. Next action: Confirm the recommendation in an independent run.

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ANOVA for factorial designs

Degrees

- of freedom follow effects

mean

- square divides SS by df

F

- and p assess evidence

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DECISION LENS · LO2

ANOVA for factorial designs — Decision Lens

1. **Meaning**
   Test model terms against residual noise

2. **Evidence**
   Identify the observation that would change the decision.

3. **Guardrail**
   Report effect size and confidence intervals alongside p-values

4. **Follow-up**
   Confirm the recommendation independently.

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The 2^3 cube

1. Eight — corners
2. three — main effects
3. three — two-way interactions
4. one — three-way interaction

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DECISION LENS · LO2

The 2^3 cube — Decision Lens

1. Interpret the effect
2. Check assumptions
3. Translate to engineering units
4. Confirm independently

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General 2^k designs

STRUCTURE

1. 2^k treatment combinations
2. orthogonal effect estimates

EVIDENCE

1. effect hierarchy guides model reduction
2. cost grows exponentially

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General 2^k designs — Decision Lens

Interpret

- Run count doubles with every factor
- State the evidence needed.

Control

- Use fractional designs when screening many factors
- Check model assumptions.

Act

- Translate to operating units.
- Confirm in an independent run.

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Pareto and normal effect plots

Standardised

- effects show magnitude

reference

- line guides screening

normal

- plot separates signal from noise

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Pareto and normal effect plots — Decision Lens

INTERPRET

1. Prioritise candidate effects
2. Evidence question: What observation would change the current decision?
3. Model risk: Check assumptions before interpreting the effect.

APPLY

1. Plots support, but do not replace, model assumptions
2. Engineering use: Translate coded results back to operating units.
3. Next action: Confirm the recommendation in an independent run.

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Centre points and curvature

Centre

- runs estimate local curvature

replicated

- centres provide pure error

significant

- curvature motivates RSM

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DECISION LENS · LO2

Centre points and curvature — Decision Lens

1

Meaning

Check whether a straight plane is adequate

2

Evidence

Identify the observation that would change the decision.

3

Guardrail

A centre point does not identify which quadratic term matters

4

Follow-up

Confirm the recommendation independently.

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Replication strategy

1

Independent — replicates capture process variation

2

duplicate — measurements capture measurement variation

3

power — increases with information

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Replication strategy — Decision Lens

1. Interpret the effect
2. Check assumptions
3. Translate to engineering units
4. Confirm independently

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Blocking

STRUCTURE

1. Group homogeneous runs
2. randomise within blocks

EVIDENCE

1. estimate block effect
2. preserve treatment balance

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Blocking — Decision Lens

Interpret

- Remove known nuisance variation
- State the evidence needed.

Control

- A block should not be confused with a factor of scientific interest
- Check model assumptions.

Act

- Translate to operating units.
- Confirm in an independent run.

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Confounding in blocks

1. **Choose**
   defining contrast
2. **aliases**
   follow
3. **high-order**
   terms are typical sacrifices
4. **document**
   the alias structure

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Confounding in blocks — Decision Lens

INTERPRET

1. Trade an interaction for practical run groups
2. Evidence question: What observation would change the current decision?
3. Model risk: Check assumptions before interpreting the effect.

APPLY

1. Never confound an effect that stakeholders consider critical
2. Engineering use: Translate coded results back to operating units.
3. Next action: Confirm the recommendation in an independent run.

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TOPIC 02 · HANDS-ON

Reduce Order-Processing Time with a 2^2 Factorial

LAB 3

A distribution centre tests Current/New ordering systems and Packing A/B to minimise processing hours.

EXPECTED OUTPUT

A 12-run replicated design, effect table, interaction interpretation and recommended setting.

TOOLS NovaDOE · NovaDOE full factorial · Excel/CSV

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Reduce Order-Processing Time with a 2^2 Factorial

ACCEPTANCE CHECK

All four combinations have three independent replicates and the recommendation cites both the predicted response and interaction risk.

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TOPIC 02 · HANDS-ON

Model a 2^3 Composite Bonding Experiment

LAB 4

A bonding team studies temperature, pressure and surface preparation to maximise joint strength.

EXPECTED OUTPUT

An eight-corner cube design, effect hierarchy, coded regression and predicted optimum.

TOOLS NovaDOE · NovaDOE full factorial · Excel/CSV

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Model a 2^3 Composite Bonding Experiment

ACCEPTANCE CHECK

The workbook contains all seven effect estimates, a hierarchical model, residual check and a recommendation in engineering units.

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TOPIC 02 · HANDS-ON

Block a Replicated Seal-Strength Experiment

LAB 5

Seal trials span morning and afternoon shifts whose ambient conditions differ.

EXPECTED OUTPUT

Two balanced blocks, randomisation within block, block-adjusted effects and deviation log.

TOOLS NovaDOE · NovaDOE full factorial + blocked run sheet · Excel/CSV

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LAB 5 · ACCEPTANCE

Block a Replicated Seal-Strength Experiment

ACCEPTANCE CHECK

Each block contains all four treatments, run order is randomised within block and the block-adjusted conclusion is documented.

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TOPIC RECAP

Recap — Factorial Design of Experiments

1

Knowledge

Factorial design and related principles.

2

Ability

Estimate two main effects and their interaction from a replicated 2^2 experiment.

3

Evidence

A NovaDOE export, validated workbook and engineering interpretation.

4

Next

Use the evidence to narrow, augment or confirm the design.

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End of Day 1

Return at 9:30 AM for Day 2

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03

TOPIC 03

Fractional Factorial and Screening Designs

Generators, aliases, resolution, Plackett-Burman and sequential experimentation

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ALIGNMENT · LO3

Key Concepts — Fractional Factorial and Screening Designs

1. **Why fractionate**
   Screen many factors with fewer runs

2. **One-half fractions**
   Use a generator to select half the full design

3. **Alias structure**
   Know which effects are inseparable

4. **Resolution III**
   Main effects alias with two-factor interactions

5. **Resolution IV**
   Main effects clear of two-factor interactions

6. **Resolution V**
   Main effects and two-factor interactions are separately estimable

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Why fractionate

1. **Assume**
   effect sparsity
2. **select**
   a defensible fraction
3. **preserve**
   critical estimates
4. **reserve**
   budget for confirmation

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Why fractionate — Decision Lens

INTERPRET

1. Screen many factors with fewer runs
2. Evidence question: What observation would change the current decision?
3. Model risk: Check assumptions before interpreting the effect.

APPLY

1. A fraction is a strategic information trade, not free efficiency
2. Engineering use: Translate coded results back to operating units.
3. Next action: Confirm the recommendation in an independent run.

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One-half fractions

Generator

- defines included runs

defining

- relation creates aliases

signs

- determine fraction

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One-half fractions — Decision Lens

1. **Meaning**
   Use a generator to select half the full design

2. **Evidence**
   Identify the observation that would change the decision.

3. **Guardrail**
   Record the generator so the design is reproducible

4. **Follow-up**
   Confirm the recommendation independently.

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Alias structure

1

Multiply — effects by defining words

2

aliases — share estimates

3

resolution — summarises shortest word

4

subject — knowledge breaks ambiguity

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Alias structure — Decision Lens

1. Interpret the effect
2. Check assumptions
3. Translate to engineering units
4. Confirm independently

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Resolution III

STRUCTURE

1. Useful for early screening
2. smallest run count

EVIDENCE

1. strong interaction risk
2. confirmation essential

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Resolution III — Decision Lens

Interpret

- Main effects alias with two-factor interactions
- State the evidence needed.

Control

- Accept only when interactions are plausibly negligible
- Check model assumptions.

Act

- Translate to operating units.
- Confirm in an independent run.

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Resolution IV

Main

- effects alias with three-factor interactions

two-factor

- interactions alias in pairs

common

- screening compromise

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Resolution IV — Decision Lens

INTERPRET

1. Main effects clear of two-factor interactions
2. Evidence question: What observation would change the current decision?
3. Model risk: Check assumptions before interpreting the effect.

APPLY

1. Escalate ambiguous interaction pairs in a follow-up design
2. Engineering use: Translate coded results back to operating units.
3. Next action: Confirm the recommendation in an independent run.

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Resolution V

Main

- effects alias with four-factor terms

two-factor

- interactions alias with three-factor terms

more

- runs

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Resolution V — Decision Lens

1

Meaning

Main effects and two-factor interactions are separately estimable

2

Evidence

Identify the observation that would change the decision.

3

Guardrail

Choose when interactions matter to the decision

4

Follow-up

Confirm the recommendation independently.

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One-quarter fractions

1. Two — defining words plus their product
2. alias — chains lengthen
3. resolution — depends on generator choice
4. efficiency — increases

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One-quarter fractions — Decision Lens

1. Interpret the effect
2. Check assumptions
3. Translate to engineering units
4. Confirm independently

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Foldover designs

STRUCTURE

1. Reverse selected factor signs
2. combine original and foldover

EVIDENCE

1. separate main effects from interactions
2. retain sequential learning

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Foldover designs — Decision Lens

Interpret

- Break targeted aliases in a follow-up
- State the evidence needed.

Control

- Plan foldover before consuming the entire budget
- Check model assumptions.

Act

- Translate to operating units.
- Confirm in an independent run.

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Plackett-Burman screening

1. **N**
   runs screen up to N-1 factors

2. **orthogonal**
   main-effect columns

3. **interactions**
   are not independently estimable

4. **good**
   for factor reduction

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Plackett-Burman screening — Decision Lens

INTERPRET

1. Estimate many main effects economically
2. Evidence question: What observation would change the current decision?
3. Model risk: Check assumptions before interpreting the effect.

APPLY

1. Use results to design a factorial confirmation, not to optimise directly
2. Engineering use: Translate coded results back to operating units.
3. Next action: Confirm the recommendation in an independent run.

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Screening metrics

Effect

- magnitude

signal-to-noise

-

cost

- per run

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Screening metrics — Decision Lens

1. **Meaning**
   Select projects and factors using business evidence

2. **Evidence**
   Identify the observation that would change the decision.

3. **Guardrail**
   A statistically active factor may still be operationally irrelevant

4. **Follow-up**
   Confirm the recommendation independently.

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Sequential experimentation



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Sequential experimentation — Decision Lens

1. Interpret the effect
2. Check assumptions
3. Translate to engineering units
4. Confirm independently

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Design review and run sheet

STRUCTURE

1. Scope
2. owners

EVIDENCE

1. factor table
2. randomised run order

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Design review and run sheet — Decision Lens

Interpret

- Make the experiment executable
- State the evidence needed.

Control

- Freeze the run sheet before execution and log deviations
- Check model assumptions.

Act

- Translate to operating units.
- Confirm in an independent run.

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Screen Five Factors with a Resolution V Half Fraction

LAB 6

A polymer extrusion team has five controllable factors but only 16 safe pilot runs.

EXPECTED OUTPUT

A 16-run fraction, generator and alias record, screened effects and confirmation plan.

TOOLS NovaDOE · NovaDOE fractional factorial · Excel/CSV

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Screen Five Factors with a Resolution V Half Fraction

ACCEPTANCE CHECK

The design has 16 unique runs, five factor columns, a stated resolution/alias assumption and a confirmation plan that does not overclaim aliased effects.

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Screen Eleven Battery-Process Factors

LAB 7

A battery lab must screen eleven formulation and process factors in twelve runs.

EXPECTED OUTPUT

A 12-run PB matrix, main-effect ranking, risk register and factorial follow-up proposal.

TOOLS NovaDOE · NovaDOE Plackett-Burman · Excel/CSV

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Screen Eleven Battery-Process Factors

ACCEPTANCE CHECK

All eleven main effects are ranked and the report explicitly states that interactions are not separately estimable in the screening design.

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TOPIC RECAP

Recap — Fractional Factorial and Screening Designs

1

Knowledge

Why fractionate and related principles.

2

Ability

Generate a defensible 2^(5-1) design, interpret aliases and plan confirmation.

3

Evidence

A NovaDOE export, validated workbook and engineering interpretation.

4

Next

Use the evidence to narrow, augment or confirm the design.

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04

TOPIC 04

Screening, Modelling and Optimising DoE

Response surfaces, CCD, Box-Behnken, mixtures and Taguchi robustness

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ALIGNMENT · LO4

Key Concepts — Screening, Modelling and Optimising DoE

1. **Response Surface Methodology**
   Model curvature after important factors are known

2. **Quadratic models**
   Curvature needs squared terms

3. **Central Composite Design**
   Augment factorial points with stars and centres

4. **CCD variants and alpha**
   Choose geometry for feasibility and prediction

5. **Box-Behnken design**
   Estimate quadratic models without all-factor extremes

6. **Contour and surface plots**
   Translate equations into operating windows

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Response Surface Methodology

1. **Fit**
   first- and second-order terms
2. **map**
   a local region
3. **find**
   stationary behaviour
4. **optimise**
   with constraints

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Response Surface Methodology — Decision Lens

INTERPRET

1. Model curvature after important factors are known
2. Evidence question: What observation would change the current decision?
3. Model risk: Check assumptions before interpreting the effect.

APPLY

1. RSM is local—avoid extrapolation beyond the design space
2. Engineering use: Translate coded results back to operating units.
3. Next action: Confirm the recommendation in an independent run.

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Quadratic models

Linear

- terms give slope

interaction

- terms twist the surface

squared

- terms create curvature

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Quadratic models — Decision Lens

1. **Meaning**
   Curvature needs squared terms

2. **Evidence**
   Identify the observation that would change the decision.

3. **Guardrail**
   Respect hierarchy when retaining higher-order terms

4. **Follow-up**
   Confirm the recommendation independently.

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Central Composite Design

1. Factorial — corners estimate interactions
2. axial — points estimate curvature
3. centre — points estimate pure error
4. alpha — controls geometry

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Central Composite Design — Decision Lens

1. Interpret the effect
2. Check assumptions
3. Translate to engineering units
4. Confirm independently

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CCD variants and alpha

STRUCTURE

1. Circumscribed extends beyond factorial range
2. inscribed keeps axial points within bounds

EVIDENCE

1. face-centred uses alpha 1
2. rotatability stabilises variance

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CCD variants and alpha — Decision Lens

Interpret

- Choose geometry for feasibility and prediction
- State the evidence needed.

Control

- Safety limits may rule out circumscribed designs
- Check model assumptions.

Act

- Translate to operating units.
- Confirm in an independent run.

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Box-Behnken design

1. **Mid-edge** treatment combinations
2. **centre** points
3. **at** least three factors
4. **fewer** dangerous corner runs

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Box-Behnken design — Decision Lens

INTERPRET

1. Estimate quadratic models without all-factor extremes
2. Evidence question: What observation would change the current decision?
3. Model risk: Check assumptions before interpreting the effect.

APPLY

1. Compare run count and feasible region with CCD
2. Engineering use: Translate coded results back to operating units.
3. Next action: Confirm the recommendation in an independent run.

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Contour and surface plots

Contours

- show equal-response lines

surfaces

- show curvature

overlay

- multiple requirements

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Contour and surface plots — Decision Lens

1. **Meaning**
   Translate equations into operating windows

2. **Evidence**
   Identify the observation that would change the decision.

3. **Guardrail**
   A persuasive plot still needs uncertainty and validation

4. **Follow-up**
   Confirm the recommendation independently.

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Optimisation and confirmation

1. Define — target and bounds
2. optimise — desirability
3. inspect — sensitivity
4. run — confirmation trials
5. compare — predicted and observed

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Optimisation and confirmation — Decision Lens

1. Interpret the effect
2. Check assumptions
3. Translate to engineering units
4. Confirm independently

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Mixture designs

STRUCTURE

1. Changing one component forces others to change
2. simplex geometry replaces a cube

EVIDENCE

1. blends drive the response
2. process variables can be added

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Mixture designs — Decision Lens

Interpret

- Component proportions sum to one
- State the evidence needed.

Control

- Ordinary factorial independence does not apply to proportions
- Check model assumptions.

Act

- Translate to operating units.
- Confirm in an independent run.

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Simplex centroid and extreme vertices

1. **Vertices**
   are pure blends

2. **edges**
   are binary blends

3. **centroid**
   is balanced blend

4. **constraints**
   reshape the feasible region

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Simplex centroid and extreme vertices — Decision Lens

INTERPRET

1. Cover constrained formulation space
2. Evidence question: What observation would change the current decision?
3. Model risk: Check assumptions before interpreting the effect.

APPLY

1. Document lower and upper component bounds
2. Engineering use: Translate coded results back to operating units.
3. Next action: Confirm the recommendation in an independent run.

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Taguchi robust design

Separate

- control and noise factors

use

- orthogonal arrays

calculate

- signal-to-noise ratio

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DECISION LENS · LO4

Taguchi robust design — Decision Lens

1

Meaning

Reduce sensitivity to uncontrollable noise

2

Evidence

Identify the observation that would change the decision.

3

Guardrail

Robustness is performance across noise, not merely a high mean

4

Follow-up

Confirm the recommendation independently.

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Signal-to-noise ratios

1

Smaller-the-better —

2

larger-the-better —

3

nominal-the-best —

4

logarithmic — ratio combines mean and variation

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Signal-to-noise ratios — Decision Lens

1. Interpret the effect
2. Check assumptions
3. Translate to engineering units
4. Confirm independently

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Evaluate and recommend

STRUCTURE

1. Compare objective to achieved effect
2. check uncertainty and residuals

EVIDENCE

1. document limitations
2. assign actions

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Evaluate and recommend — Decision Lens

Interpret

- Close the experimental learning loop
- State the evidence needed.

Control

- Recommendation strength must match evidence strength
- Check model assumptions.

Act

- Translate to operating units.
- Confirm in an independent run.

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Follow-up action portfolio

1. Confirm — promising settings
2. fold — over aliases
3. augment — for curvature
4. widen — or narrow ranges
5. improve — measurement system

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Follow-up action portfolio — Decision Lens

INTERPRET

1. Choose the next experiment intentionally
2. Evidence question: What observation would change the current decision?
3. Model risk: Check assumptions before interpreting the effect.

APPLY

1. Capture decision, owner, due date and acceptance test
2. Engineering use: Translate coded results back to operating units.
3. Next action: Confirm the recommendation in an independent run.

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TOPIC 04 · HANDS-ON

Design a Robust Golf-Ball Process with Taguchi Logic

LAB 8

A golf-equipment team tests four two-level control factors under Driver and 5-Iron noise conditions.

EXPECTED OUTPUT

An orthogonal eight-run control matrix, two noise responses, S/N analysis and robustness recommendation.

TOOLS NovaDOE · NovaDOE full/fractional matrix used as an orthogonal-array proxy · Excel/CSV

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Design a Robust Golf-Ball Process with Taguchi Logic

ACCEPTANCE CHECK

The recommended settings are justified by S/N ratio and mean response, with a confirmation plan across both club conditions.

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TOPIC 04 · HANDS-ON

Optimise Sterile-Package Seal Strength with CCD

LAB 9

A medical-packaging engineer targets seal strength of 26 N within an acceptable 20–32 N range.

EXPECTED OUTPUT

A CCD export, quadratic model table, contour evidence, optimum and confirmation record.

TOOLS NovaDOE · NovaDOE Central Composite · Excel/CSV

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Optimise Sterile-Package Seal Strength with CCD

ACCEPTANCE CHECK

The design includes factorial, axial and centre points; the recommendation targets 26 N and includes confirmation evidence.

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Optimise a Three-Component Thermal Paste Mixture

LAB 10

A thermal-materials lab blends ceramic filler, binder and additive; proportions must total 100%.

EXPECTED OUTPUT

A constrained candidate set, simplex plot table, blend model evidence and validation plan.

TOOLS NovaDOE · NovaDOE Latin Hypercube for candidate exploration + simplex filtering · Excel/CSV

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Optimise a Three-Component Thermal Paste Mixture

ACCEPTANCE CHECK

Every retained blend sums to 1.000 within tolerance, satisfies bounds and leads to a constrained follow-up recommendation.

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TOPIC RECAP

Recap — Screening, Modelling and Optimising DoE

1

Knowledge

Response Surface Methodology and related principles.

2

Ability

Separate control and noise factors and recommend robust rather than merely high-mean settings.

3

Evidence

A NovaDOE export, validated workbook and engineering interpretation.

4

Next

Use the evidence to narrow, augment or confirm the design.

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05

WRAP-UP

Evidence, Feedback & Next Steps

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Course Feedback

1

Share your feedback

Complete the course feedback form before you leave.

2

What worked

Tell us which labs and concepts helped you most.

3

What to improve

Suggest where you needed more time or examples.

4

Certificate

Your certificate of completion is emailed after the course.

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THANK YOU

Design fewer experiments. Learn more from every run.

Control variation, expose interactions and confirm recommendations before release.

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